

Course Code: EET201

Course Name: CIRCUITS AND NETWORKS

Max. Marks: 100

Duration: 3 Hours

PART A

Answer all questions. Each question carries 3 marks

Marks

- 1 Using superposition, find V_o in the circuit of Fig. 1

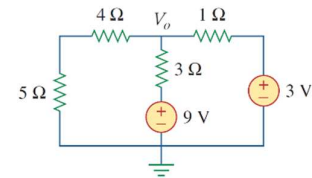


Fig.1

- 2 Find the Norton equivalent of the circuit in Fig.2 as seen by looking into terminals a and b.

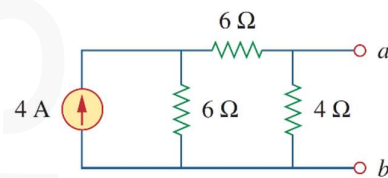


Fig.2

- 3 Obtain the expression for current in the circuit shown in fig.3 for $t > 0$, assume no initial charge across capacitor. Use Laplace transform method. V_s is a constant DC voltage source

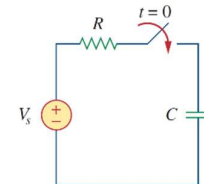


Fig.3

- 4 Explain how to obtain s-domine equivalent of Resistance, inductance and capacitance
- 5 Explain the use of dot convention in the analysis of coupled circuits.
- 6 Find $v(t)$, for $t > 0$ in the circuit of Fig.5
 $V_s = 20$ V.

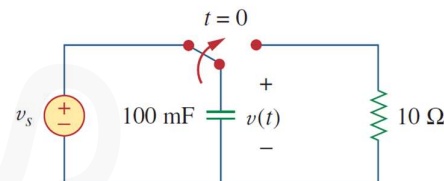


Fig.5

- 7 Derive the expression for resonance frequency in a series RLC circuit connected to a variable frequency AC source
- 8 An unbalanced Y-load $Z_A = 15 \Omega$, $Z_B = 10 + j5 \Omega$, $Z_C = 6 - j8 \Omega$. is connected to a 100 V, 3phase, 4wire, ACB sequence system. Calculate the line currents

- and the neutral current.
- 9 Obtain T parameters in terms of Z parameters (3)
- 10 Show that the overall Y-parameter matrix of two parallel connected 2-port networks is the sum of the Y-parameter matrices of the individual networks. (3)

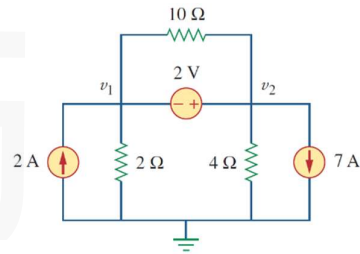
PART B

Answer any one full question from each module. Each question carries 14 marks

Module 1

- 11(a) For the circuit shown in Fig.6, find the node voltages v_1 and v_2

Fig.6



(10)

- (b) Use source transformation to reduce the circuit in Fig.7 to a single voltage source in series with a single resistor.

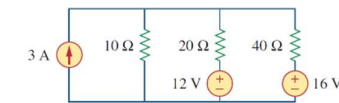


Fig.7

(4)

- 12 Use mesh analysis to evaluate I_o in the circuit in fig.8

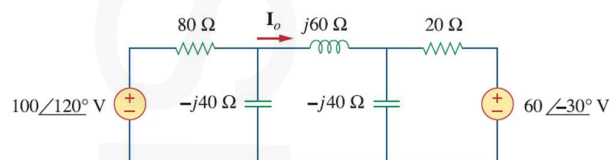


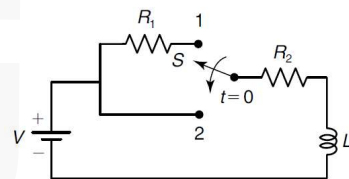
Fig.8

(14)

Module 2

- 13 In the circuit shown in Fig.9, the switch S has been in position 1 for a long period of time. Find the complete expression for the current after throwing the switch S to position 2

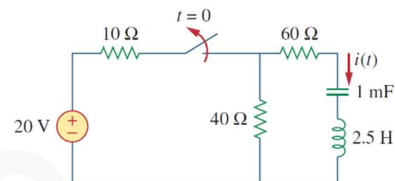
Fig.9



(14)

- 14 Find $i(t)$ for $t > 0$ in the circuit of Fig.10

Fig.10

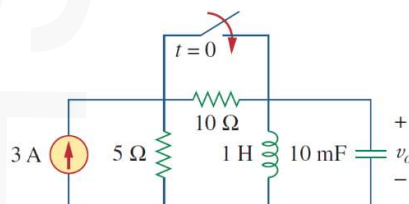


(14)

Module 3

- 15 Initially the circuit is in steady state and the switch is closed at time $t=0$. Find the output voltage $v_o(t)$ for $t>0$ in the circuit of fig.11

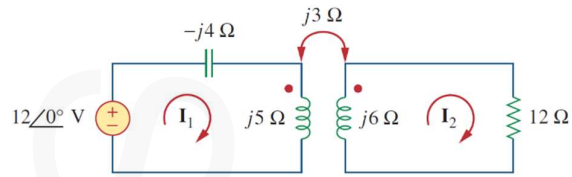
Fig.11



(14)

- 16 Calculate the phasor currents

I_1 and I_2 in the circuit of Fig.12



(14)

Fig.12

Module 4

- 17 A star connected load of $Z_a=80 + j50 \Omega$, $Z_b=20 + j30 \Omega$, $Z_c=60 - j40 \Omega$ is connected to a 110V, 3phase 3 wire system. Determine the line currents . (14)

- 18(a) Find the phase currents and line currents in the circuit in fig.14. $Z_1=8+j6 \Omega$, $Z_2=4.2-j2.3 \Omega$, $Z_3=10 \Omega$ (9)

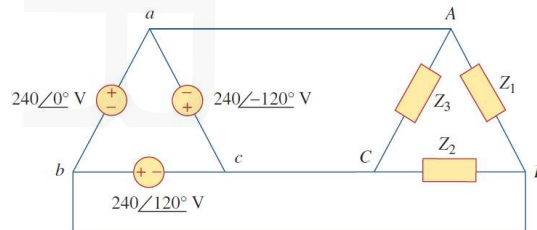
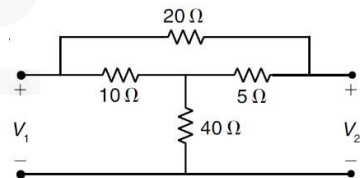


Fig.14

- (b) Derive the expression for half power frequencies in case of a series RLC circuit (6)

Module 5

- 19(a) Calculate the Y parameters of the network in fig.15 considering as two networks connected in parallel



(10)

Fig.15

- (b) Express T parameters in terms of Y parameters (4)

- 20(a) Calculate the T parameters of the network shown in fig.16 (8)

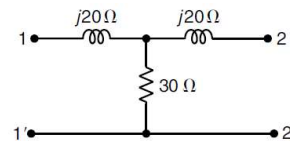


Fig.16

- (b) Drive the condition for reciprocity and symmetry of two port network in terms of h parameters (6)